Lab Section #9: Hypothesis Testing

Today's Content

Homework 6 Review

Review Concepts from Lecture

Using Posit Cloud for Hypothesis Testing

Discussion for Homework 7

Review: Confidence Intervals

What is a confidence interval?

Confidence interval is an *interval* estimate of a population parameter. It gives a range around a sample statistic.

When do we use confidence Intervals?

We use confidence intervals when we want to estimate a population value using data from a sample. For example, we might survey 100 college students to estimate the average number of hours all college students spend studying.

What do you need to calculate before you construct a confidence interval?

We need to calculate the standard error (SE). It measures how much a sample statistic, such as a sample mean, is expected to deviate from the true population mean. We also need the sample statistic and an appropriate critical value based on the confidence level.

Review: Confidence Intervals

How do you construct a confidence interval?

The general formula is:

Confidence Interval = [sample mean or percentage - (Z-score x SE), sample mean or percentage + (Z-score x SE)

How do you interpret the results?

Suppose we calculated a 95% confidence interval of [$24,000, $25,980] for the average starting salary. Then we can say, *"We are 95% confident that the true average starting salary of US college graduates lies between $24,020 and $25,980."*

Review: Confidence Intervals

1) Based on the output and your knowledge of the GSS survey, can we make statistical inferences about the US adult population using this sample?

→Is this a 1) large and 2) random sample?

2) Based on the output, what is the sample mean of the average hours Americans worked per week? Please calculate the standard error of the sample mean.

Sample standard deviation

$s = 10.88$

Use the formula

$SE = \frac{s}{\sqrt{n}}$

Substitute the values

$SE = \frac{10.88}{\sqrt{1{,}000}}$

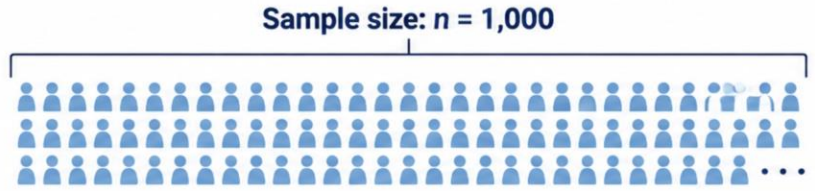
Calculate the square root

$\sqrt{1{,}000} \approx 31.62$

Standard Error

$SE = \frac{10.88}{31.62} \approx 0.34$

1 First take the square root of the sample size, then divide the standard deviation by that result.





Standard error: how much sample means typically vary from sample to sample.



Sample Mean = 42.72

Review: Confidence Intervals

3) Construct the 95% Confidence Interval and interpret the results.

1 Identify the values

Sample mean ($\bar{x}$) = **42.72**
Z critical value (for 95%) = **1.96**
Standard error (SE) = **0.34**

2 Calculate the margin of error

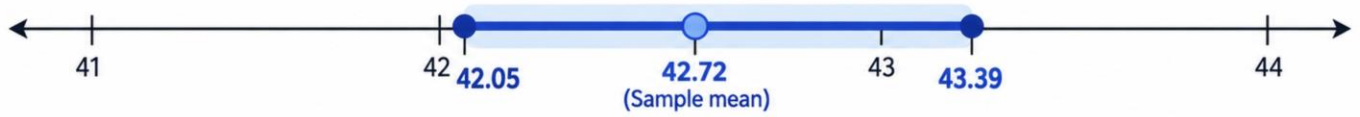
Margin of error (ME) = Z × SE
= 1.96 × 0.34
≈ **0.67**

3 Construct the confidence interval

95% CI = **42.72 ± 0.67**
Lower bound:
42.72 − 0.67 = **42.05**
Upper bound:
42.72 + 0.67 = **43.39**

4 Final answer

95% CI:
(42.05, 43.39)











We are **95%** confident that the true average hours worked per week by American adults lies between **?** and **?**.

Review: Confidence Intervals

4) Based on the output, what is the sample percentage of American adults that have a gun at home? Please calculate the standard error of the sample percentage.

Standard Error of a Sample Percentage: $$SE = \sqrt{\frac{sample\%(100 - sample\%)}{n}}$$

1 Substitute the values

$$SE = \sqrt{\frac{23.10\%(100 - 23.10\%)}{1{,}000}}$$

2 Calculate (100 – 23.10%)

$$SE = \sqrt{\frac{23.10\%(76.90)}{1{,}000}}$$

3 Multiply the numerator

$$SE = \sqrt{\frac{1{,}776.39}{1{,}000}}$$

4 Divide by 1,000

$$SE = \sqrt{\frac{1{,}776.39}{1{,}000}}$$

5 Take the square root

$$SE = \sqrt{1.78} \Rightarrow SE = 1.33$$

Review: Confidence Intervals

5) Please calculate the 95% CI of the percentage of American adults that have a gun at home in 2006, then **interpret the result**.

95% CI = sample percent ± (Z × SE)

$= 23.10 \pm (1.96 \times 1.33)$

Sample percentage = 23.10%
Standard error (SE) = 1.33 percentage points
Z (for 95% confidence) = 1.96



We are **95%** confident that the true percentage of American adults who have a gun at home lies between ___?___ and ___?___.

Review: Confidence Intervals

6) You could write some R code to replicate the calculation process and cross-check your answers *if you would like*.

```r
#calculating standard error
sqrt((23.10 *(100 - 23.10)))
```

```
[1] 42.14724
```

```r
#constructing confidence interval
23.10 - (1.96*1.33) #Lower Bound
23.10 +(1.96*1.33) #Upper Bound

```

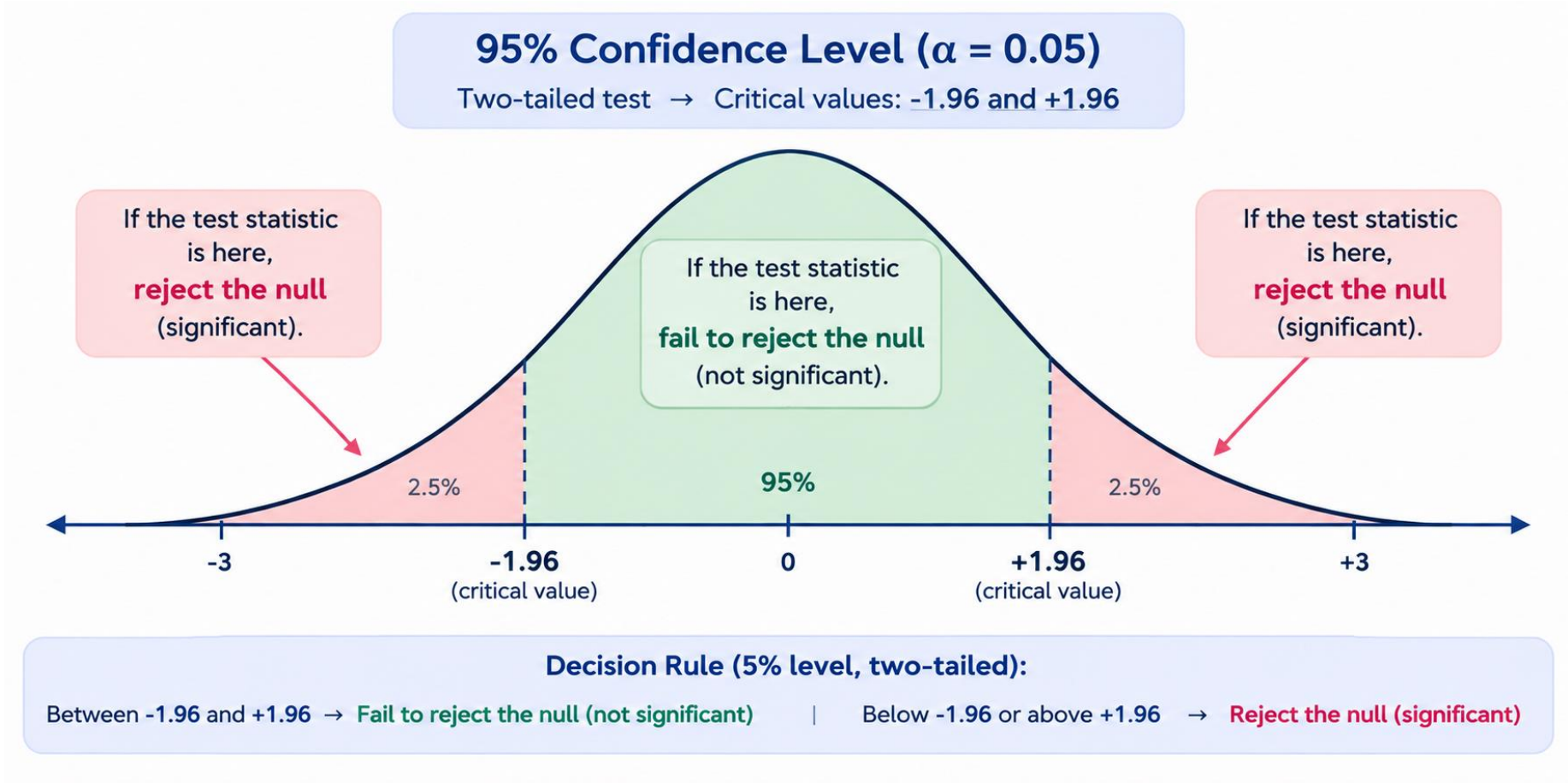
```
[1] 20.4932
[1] 25.7068
```

Review: Hypothesis Testing

Hypothesis Testing: a statistical means of testing a hypothesis about a population, based on sample data.

Hypothesis Testing (also known as significance tests) is used to determine if an observed difference between a sample statistic and a hypothesized population parameter is statistically significant or due to random chance.

Review: Test Statistics



6 Steps of Hypothesis Testing

(0) Organize your information about the hypothesized population parameter (given in the hypothesis), the sample statistic (often found in R output or calculated by yourself), and the confidence level (or the alpha level, given in HW prompts).

(1) Construct Null Hypothesis (H0) and Alternative Hypothesis (H1).

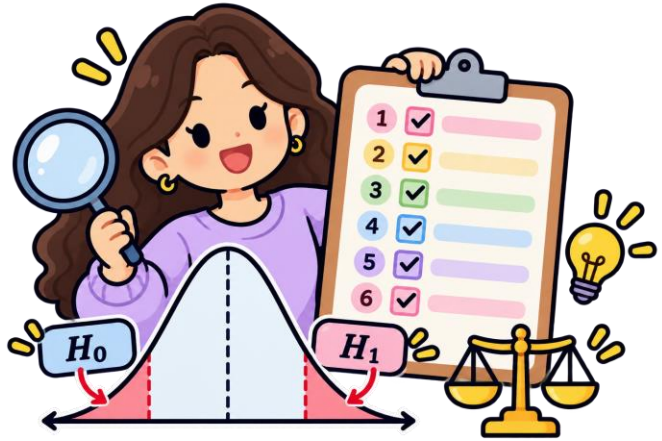
(2) Determine the critical value based on the confidence level (the Z-scores for the 95% or 99% level)

(3) Calculate the test statistic (the specific Z-score of the sample mean/proportion).

(4) Compare test statistic to the critical values: does it fall within or outside of the critical values?

(5) Make decision about whether to reject H0 based on the comparison.

(6) Interpret the results



Review: Hypothesis Testing

Be sure to organize and label all the available information about the sample and population such as the mean and standard deviation as well as the sample size and confidence level.

Step 1: State the null hypothesis and alternative hypothesis

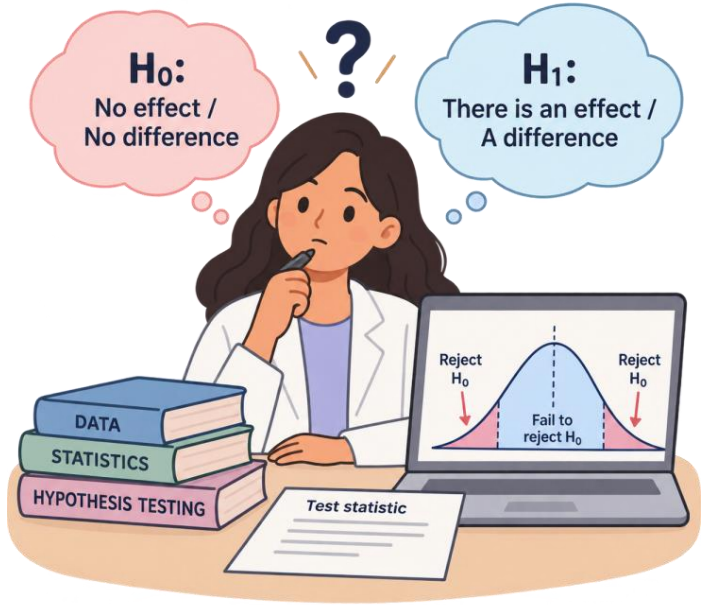
H0 Null Hypothesis: Equal

Sample Statistic = Population Statistic

Assumes no difference exists

There is no significant difference between the observed sample statistic and the hypothesized population parameter

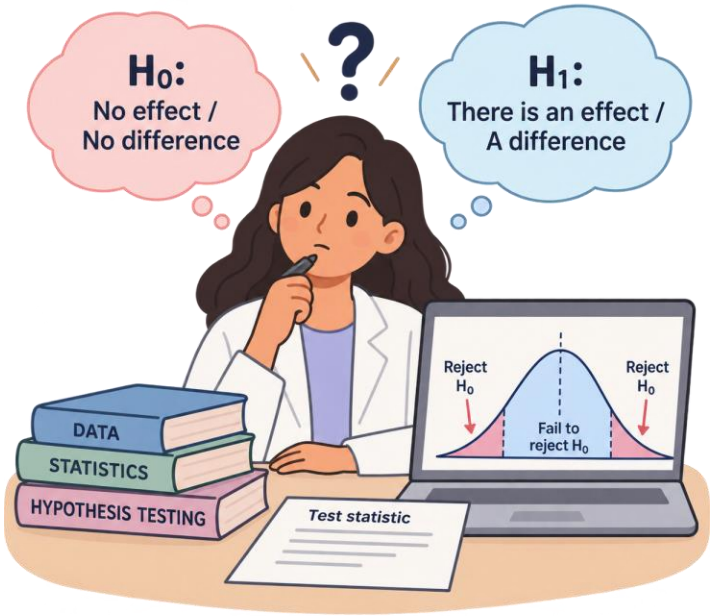
The observed difference is due to chance



Review: Hypothesis Testing

H1 Alternative Hypothesis: Not Equal

- Sample Statistic ≠ Population Statistic
- Assumes a significant difference exists
- There is a significant difference between the observed sample statistic and the hypothesized population parameter
- The observed difference is not due to chance (it is significant or real)



H0 and H1 are mutually exclusive (one is false)

We will test the H0 to draw conclusions about H1

If we fail to reject the H0 then then there is not support for the H1

If we reject the H0 then there is support for the H1

Review: Hypothesis Testing

Step 2. Determine the critical value (this is threshold for significance)

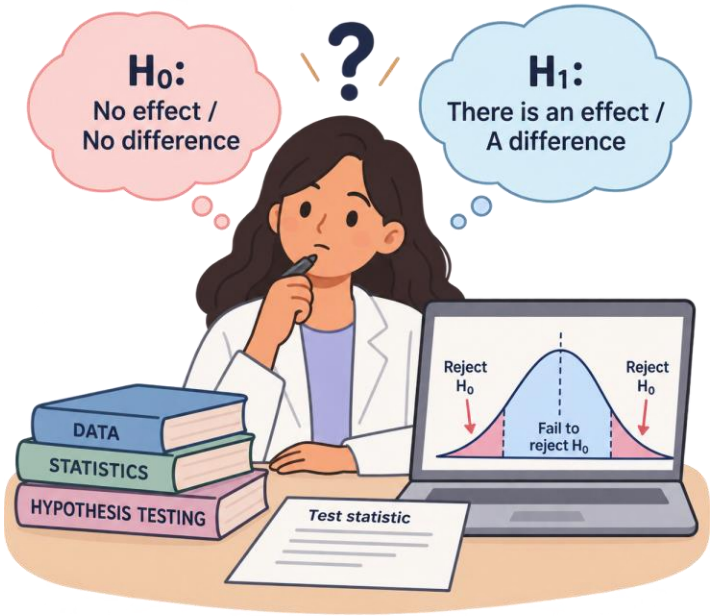
If N is ≥ 100: The critical value is a Z Score (1.96 or 2.58)

If N is < 100: The critical value is the value found on the T table

Alpha Level:

- **If 95% confidence level then alpha = 0.05, Z-score = +/- 1.96**
- **If 99% confidence level then alpha = 0.01, Z-score = +/- 2.58**

Critical values: the Z-score at the 0.05 or 0.01 alpha level



Note that, critical values take both negative and positive signs to form a critical value interval, e.g. [-1.96, 1.96], [-2.58, 2.58]

Review: Hypothesis Testing

Step 3. Calculate the test statistic

The test statistic is the value calculated with one of the formulas provided below:

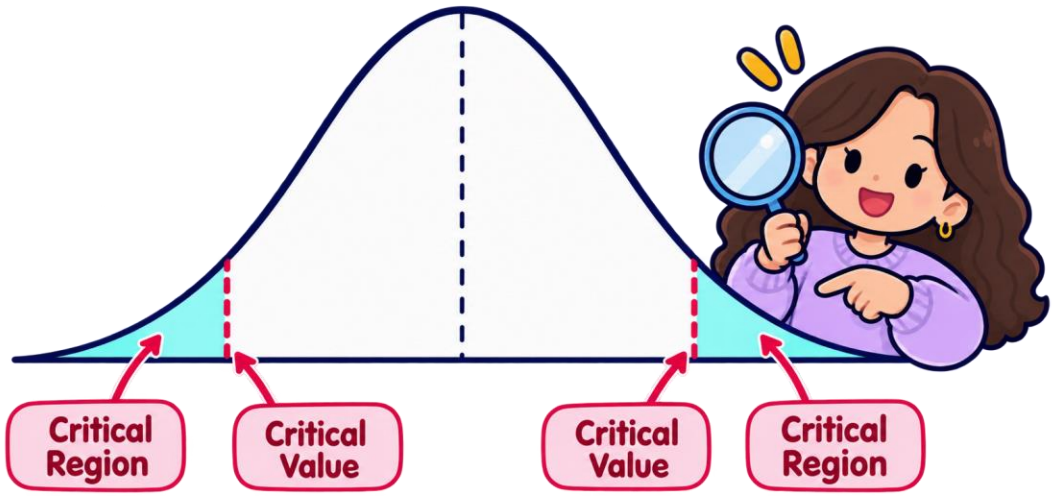
- **Use Z Score when the population SD, σ, is known**
- **Use t score when the population SD, σ, is unknown**
- **This value will be used to determine if the observed difference is significant**



Review: Hypothesis Testing

Step 4. Compare the test statistic to the critical value

Does the test statistic fall beyond either critical value, into one of the critical regions?



Critical values: The *z*-score selected in Step 2 based on the alpha level (1.96 or 2.58)

Test statistic: The *z*-score calculated in Step 3 using the appropriate formula.

Review: Hypothesis Testing

Step 5. Make a decision about the null hypothesis

Compare the test statistic with the critical values to decide whether to reject the null hypothesis.

If the test statistic falls within the critical region:

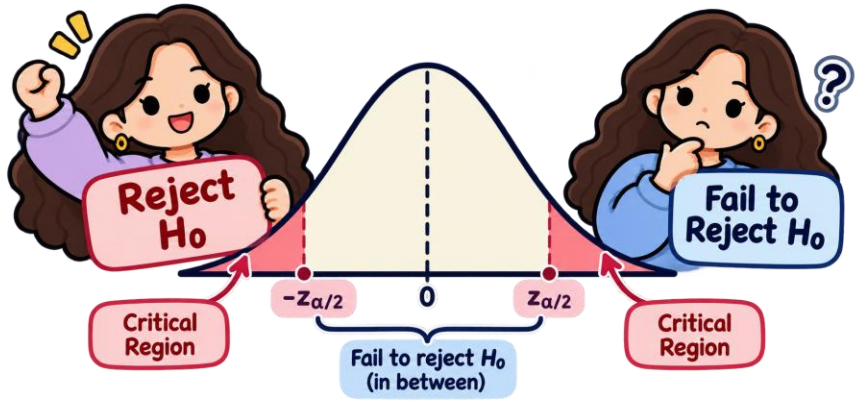
Decision: **REJECT** H_0 (Null Hypothesis)

The results provide sufficient statistical evidence to support H_1.

If the test statistic falls outside the critical region:

Decision: **FAIL TO REJECT** H_0 (Null Hypothesis)

The results do not provide sufficient statistical evidence to support H_1.



Review: Hypothesis Testing

Step 6. Interpret Results

If the test statistic falls within the critical region:

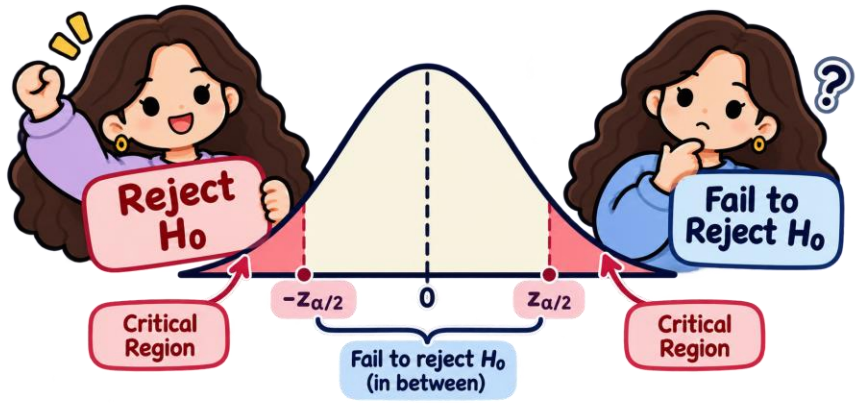
Decision: REJECT H_0 (Null Hypothesis)

Interpretation: We are 95% or 99% confident that there **is a statistically significant difference** between the sample statistic and the hypothesized population parameter.

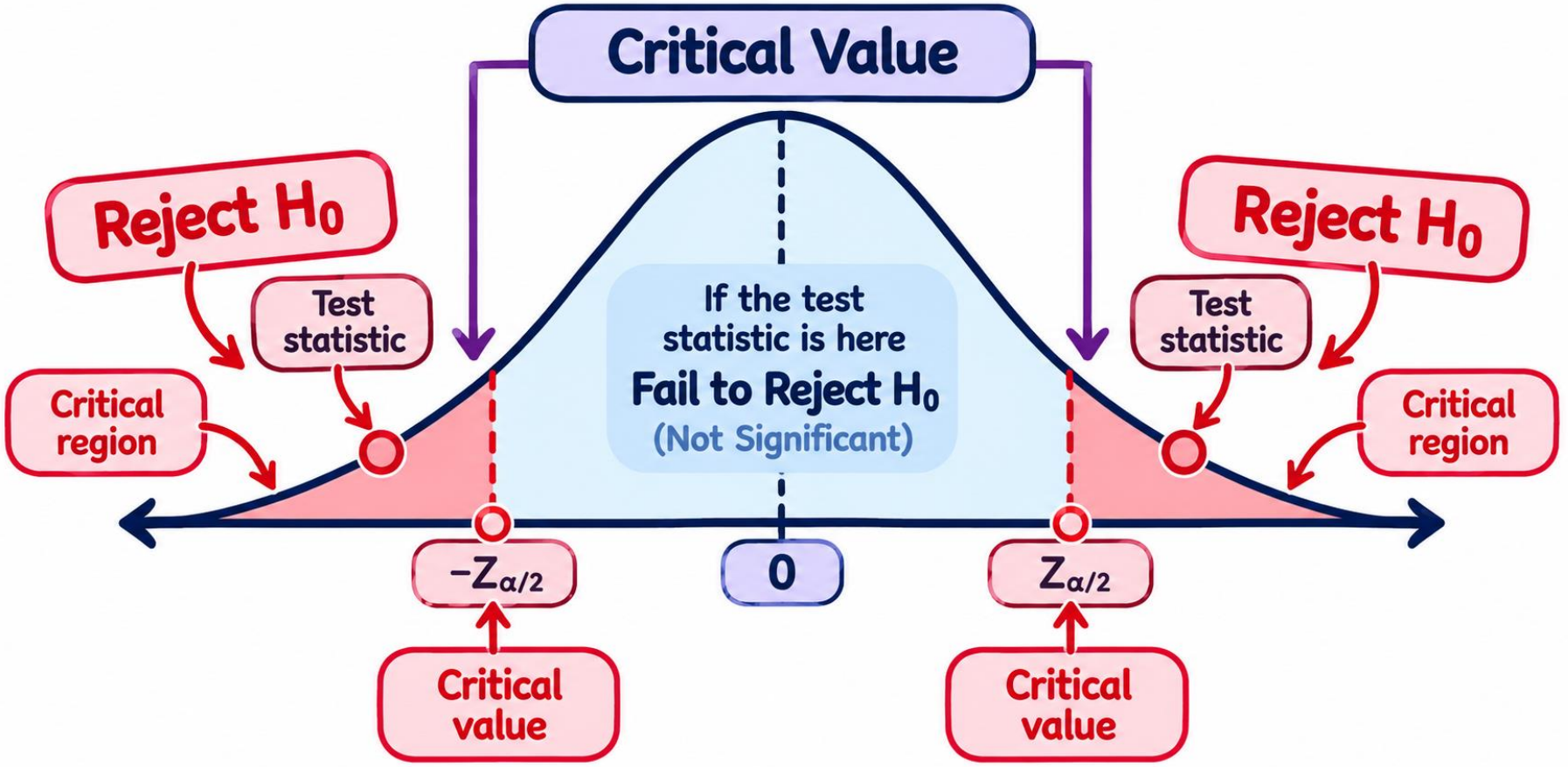
If the test statistic falls outside the critical region:

Decision: FAIL TO REJECT H_0 (Null Hypothesis)

Interpretation: We are 95% or 99% confident that **there is not a statistically significant difference** between the sample statistic and the hypothesized population parameter.

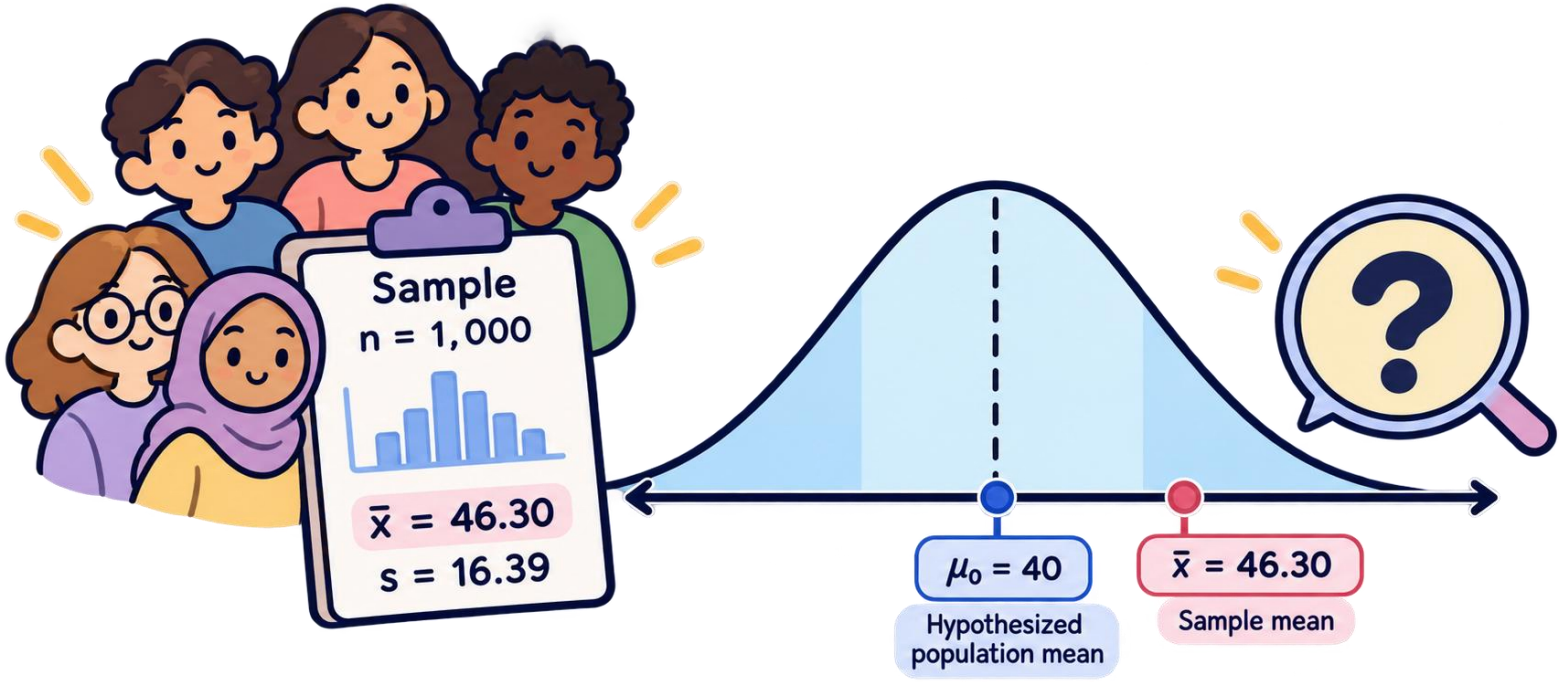


Review: Hypothesis Testing



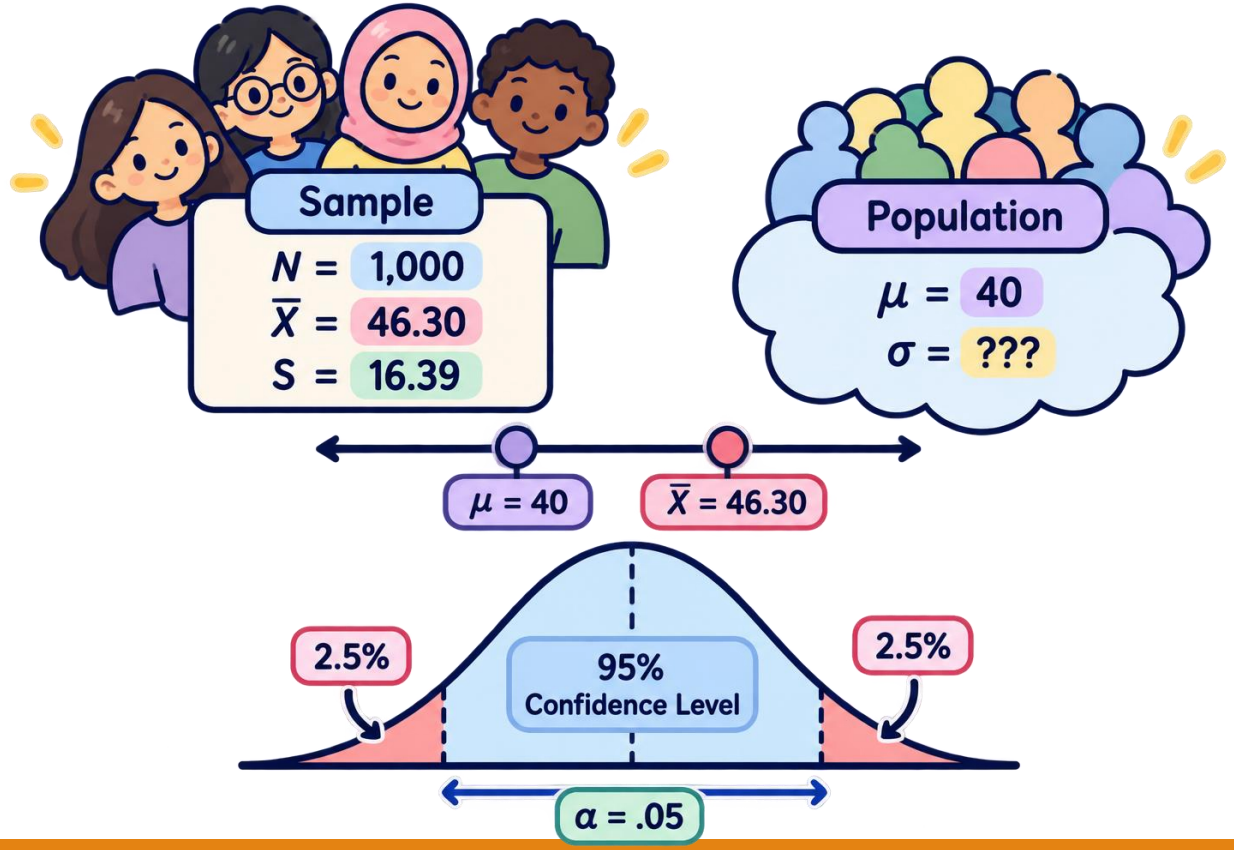
Example: Hypothesis Testing

In a sample of 1,000 respondents, the average age is 46.30 with a standard deviation of 16.39. A researcher hypothesizes that the population mean is 40. Is there a significant difference?



Example: Hypothesis Testing

(0) Organize and label all the information



Example: Hypothesis Testing

Step 1: State the null hypothesis and alternative hypothesis.

H_0 (Null Hypothesis):

There is <u>no</u> statistically significant difference between the sample mean of 46.30 years and the hypothesized population mean of 40

H_1 (Alternate Hypothesis):

There is a statistically significant difference between the sample mean of 46.30 years and the hypothesized population mean of 40



Example: Hypothesis Testing

Step 2: Determine the critical value

Do we use a Z-Score (1.96 or 2.58) or should we use the T-Table to find the critical value?

N is greater than 100 so we use a Z score. We don't need to use the T table but we could use it to double check, just to be sure.

What row would we look at on the T Table?

df= N-1 -> 1,000 -1 = 999 Bottom Row (infinity row, Z Scores)

What column do we look at on the T Table?

Number of Tails? Two Tails

Alpha Level ? 0.05

Third Column



Example: Hypothesis Testing

Step 3: Calculate the test statistic.



$$Z\ score = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}} \quad \text{or} \quad t\ score = \frac{\bar{X} - \mu}{s/\sqrt{N}} \ ?$$

Is the population standard deviation known? **No**

$$t\ score = \frac{46.30 - 40}{16.39/\sqrt{1{,}000}}$$

$$t\ score = \frac{6.30}{16.39/31.62}$$

$$t\ score = \frac{6.30}{0.52}$$

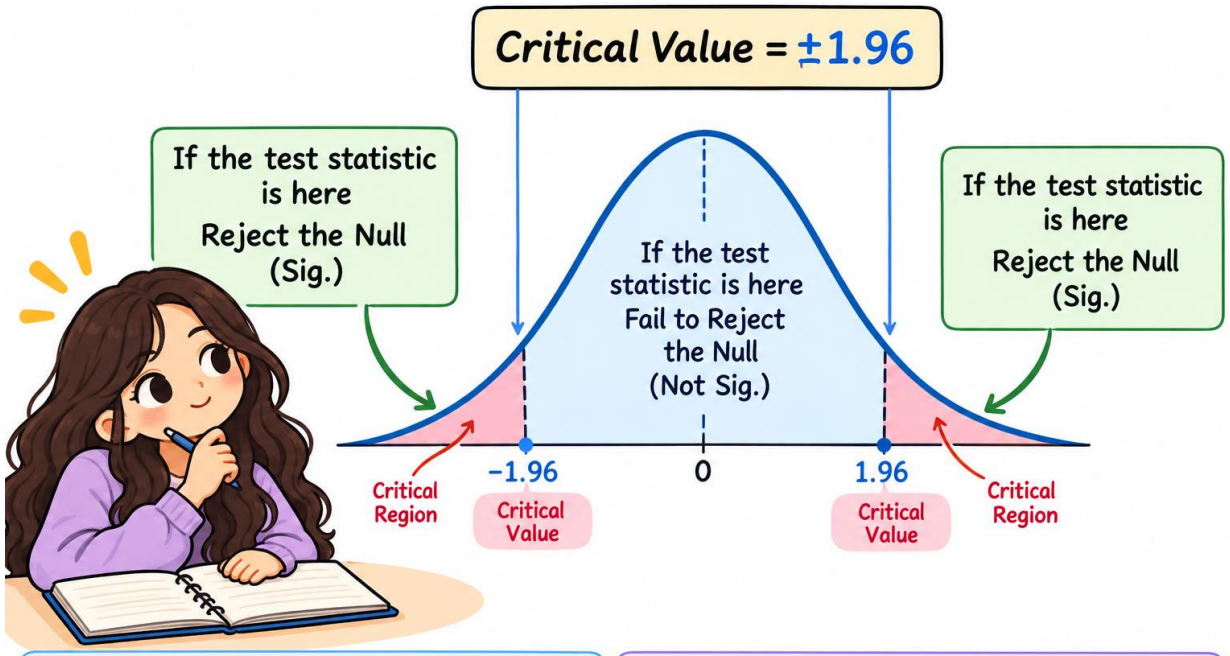
$$t\ score = 12.12$$

Test statistic: the Z-score or t-score calculated in step 3 (using the formula).

Test Statistic = 12.12

Example: Hypothesis Testing

Step 4: Compare the test statistic to the critical value.



Critical value: the Z-score or t-score selected in step 2 (using the table) = **±1.96**

Test statistic: the Z-score or t-score calculated in step 3 (using the formula) = **12.12**

Example: Hypothesis Testing

Step 5: Make a decision about the null hypothesis.

Is the test statistic (from the formula) outside the critical value interval (based on alpha level)? In other words, is the test statistic in the critical region?

If the test statistic is outside the critical value interval:

Reject the null hypothesis and we are 95% confident that a statistically significance difference **does exists.**



If the test statistic is within the critical value interval:

Fail to reject the null hypothesis and we are 95% confident that a statistically significance difference **doesn't exists.**

Decision: **The test statistic of 12.12 is outside the critical value interval of [-1.96, 1.96] so we Reject the Null Hypothesis**

Example: Hypothesis Testing

Step 6: Interpret Results

*We **Rejected** the Null Hypothesis so we conclude that we are 95% confident that there **is a statistically significant difference** between the sample average age of 46 and the hypothesized population average age of 40.*



Using R For Hypothesis Testing

You should practice how to do hypothesis testing by hand. But you can also use R to do the test.

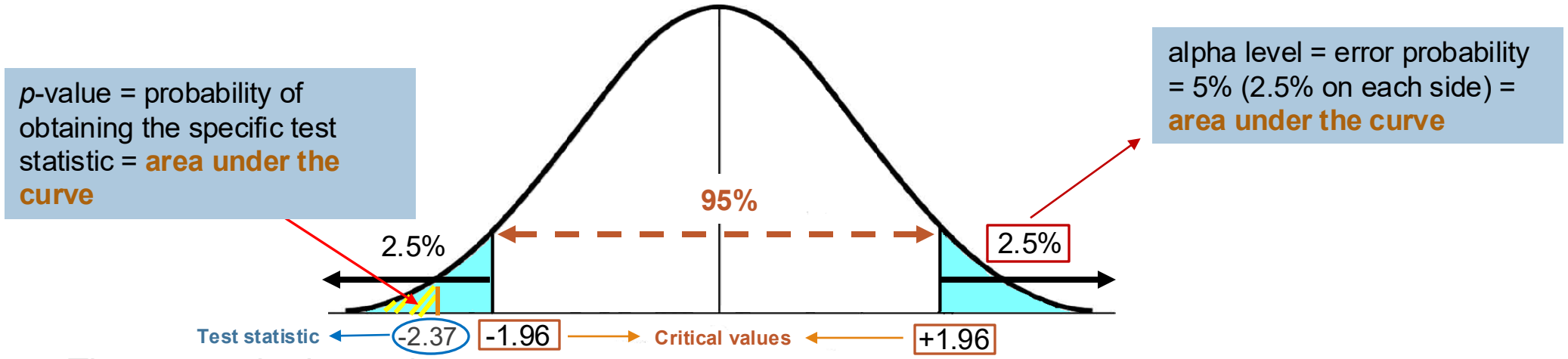
R makes the process easy –

but if you don't understand the step-by-step hand calculation process, you **will NOT be able to interpret** the R outputs.

P-value is the key to understand R outputs

Two methods of conducting hypothesis testing :

- By hand: Comparing the test statistic to the critical values (e.g., +/-1.96), both of which are Z-scores.
- Use R: comparing the p-value, i.e., probability of obtaining the sample statistic, to the alpha levels (0.05 or 0.01), both of which are probabilities (area under the curve)



  - The two methods are the same:
    - Any sample mean/percentage is associated with a specific test statistic (Z-score, hand-calculated);
    - Any sample mean/percentage is associated with a specific *p*-value (probability, R output);
  - Decisions about the null hypothesis can be made solely based on the test statistic ***or*** the *p*-value.

P-value is the key to understand R outputs

How to interpret *p*-value?

- *p*-value < 0.05 (or 0.01)

  → the probability of getting this specific sample statistic is too small (less than 5% chance) → H_0 is likely ***NOT*** true

- reject H_0

- *p*-value >= 0.05 (or 0.01)

  → it's likely to get this sample statistic (>= 5% chance) → H_0 is likely True

- fail to reject H_0 ("accept" H_0)

- **You can think of p-value as the probability of H_0 being true** (inaccurate, but convenient for understanding.)

P-value is the key to understand R outputs

When using R, you will look for the p-value in the R output.

However, R sometimes writes p-values in a way called "scientific notation."

How to read a scientific notation?

$10 = 10^{1}$, $100 = 10^{2}$, $1000 = 10^{3}$; can be written as e+1, e+2, e+3

$0.1 = 10^{-1}$, $0.01 = 10^{-2}$, $0.001 = 10^{-3}$; can be written as e-1, e-2, e-3

8.334 e-3 = 8.334×10^{-3} = 8.334×0.001 = 0.008334

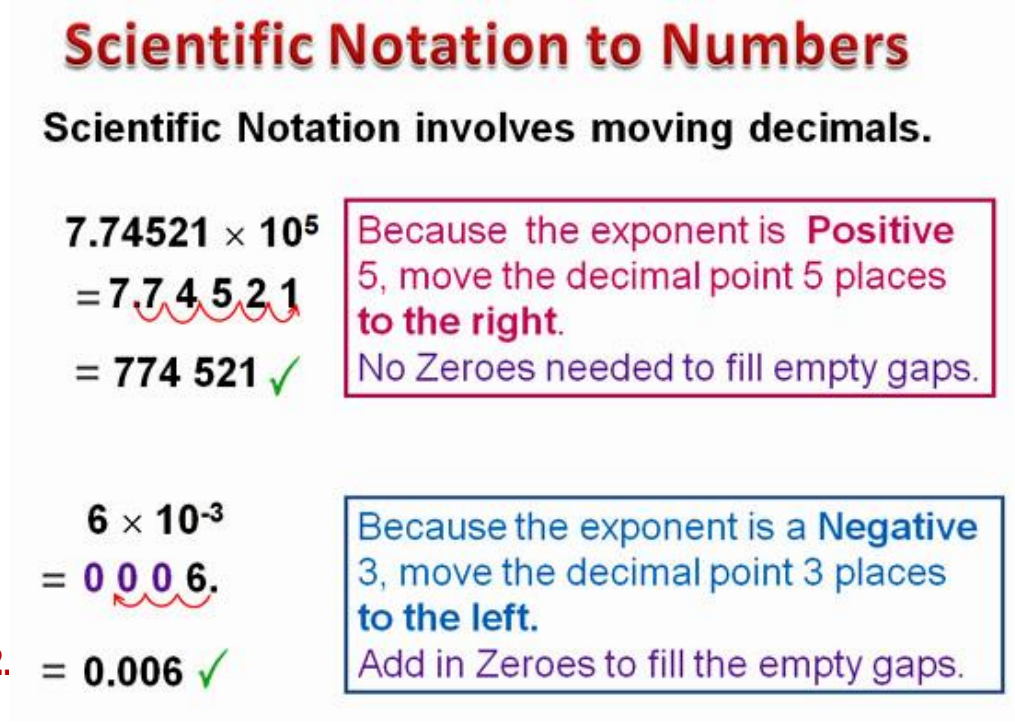
6.189 e-4 = 6.189×10^{-4} = 6.189×0.0001 = 0.0006189

0.05 = ?

0.05 = $5 \times 0.01 = 5 \times 10^{-2} = 5 \times$ e-2 = 5 e-2

0.01 = $1 \times 0.01 = 1 \times 10^{-2} = 1 \times$ e-2 = 1 e-2

any number times e-3, e-4, e-5 … will be much smaller than e-2.



R for Hypothesis Testing

```
> t.test(gss$age, mu=40)

        One Sample t-test

data: gss$age
t = 12.148, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not
equal to 40
95 percent confidence interval: 45.27813
47.31187
sample estimates:
mean of x
46.295
```

By default, R uses a **95%** confidence level.

t: test statistic t-score (z-score is a special case of t-score)
df: degree of freedom = sample size – 1
p-value: the probability of observing the sample mean if H_0 were true.
alternative hypothesis: H_1 (you need to specify H_0 by yourself.)
95 percent CI: the CI of the sample mean
sample estimates: the sample mean

H_0: $\bar{X} = \mu$, or the true population mean is 40
H_1: $\bar{X} \neq \mu$, or the true population mean is ***NOT*** 40.
p-value = 0.00000000000000022 < 0.05
decision = ***reject the null hypothesis***

Interpretation

```
> t.test(gss$age, mu=40)

        One Sample t-test

data: gss$age
t = 12.148, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not
equal to 40
95 percent confidence interval: 45.27813
47.31187
sample estimates:
mean of x
46.295
```

By default, R uses a **95%** confidence level.

The sample average age is 46.30 years old. The probability of obtaining this value if the true population average age is 40 is less than 0.0000000000000002.

In other words, we ***reject the null hypothesis*** and conclude that there is a statistically significant difference between the sample average age (46) and the hypothesized population average age (40) at the 0.05 significance (alpha) level.

Using Posit Cloud for Hypothesis Testing

When we conduct hypothesis testing by hand, we compare the test statistic to the critical values to determine whether the result falls within the critical region.

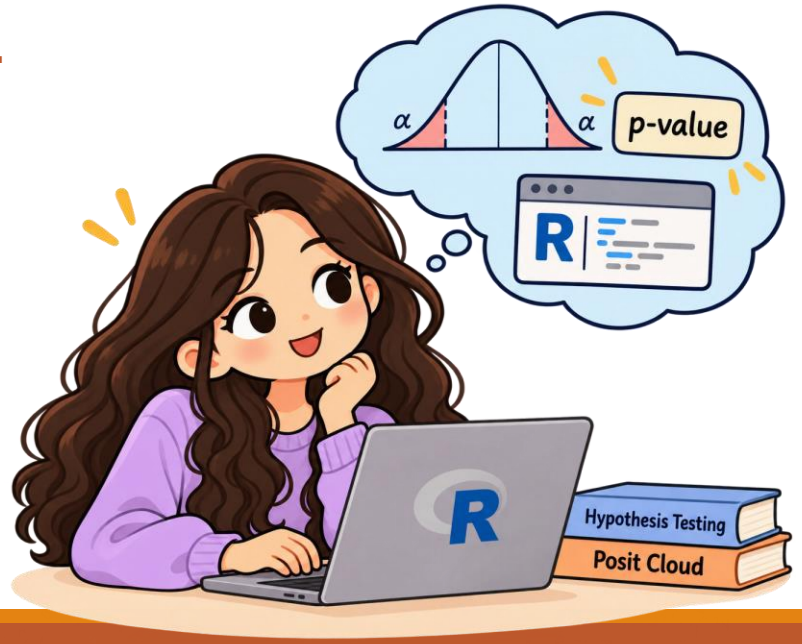
This helps us decide whether to reject the null hypothesis at our chosen significance level ($\alpha = .05$ or $\alpha = .01$).

Posit Cloud allows us to perform these calculations quickly using R.

Instead of comparing the test statistic to the critical values, we can use R to calculate a p-value.

P-value: The probability of obtaining the specific sample statistic, assuming the null hypothesis is true.

We then compare the p-value to our chosen significance level (α) to make a decision about the null hypothesis.



When Using Posit Cloud

Test Statistic: the z-score of the specific sample mean (how far it is from the hypothesized population mean)

P-value: the probability of the specific sample mean (how likely we can obtain this sample mean given the hypothesized population mean).

We will compare the p-value to the alpha level of .05 or .01 (depending on your confidence level).

You can use hand-calculated test statistics or p-values generated by R for hypothesis testing.

You can use one or the other (not both).



Posit Cloud: Hypothesis Testing

Posit Cloud will provide p-values that represents the probability of error if we reject the null hypothesis:

If the p-value is greater than 0.05:

- **It's likely to obtain this sample mean given the hypothesized population mean**
- **No significant difference between the sample mean and population parameter**
- **There is support for null hypothesis**

If the p-value is 0.05 or less:

- **It's unlikely to obtain this sample mean (at the 95% level)**
- **There is a significant difference between the sample mean and population parameter**
- **Reject the null hypothesis**

Posit Cloud: Hypothesis Testing

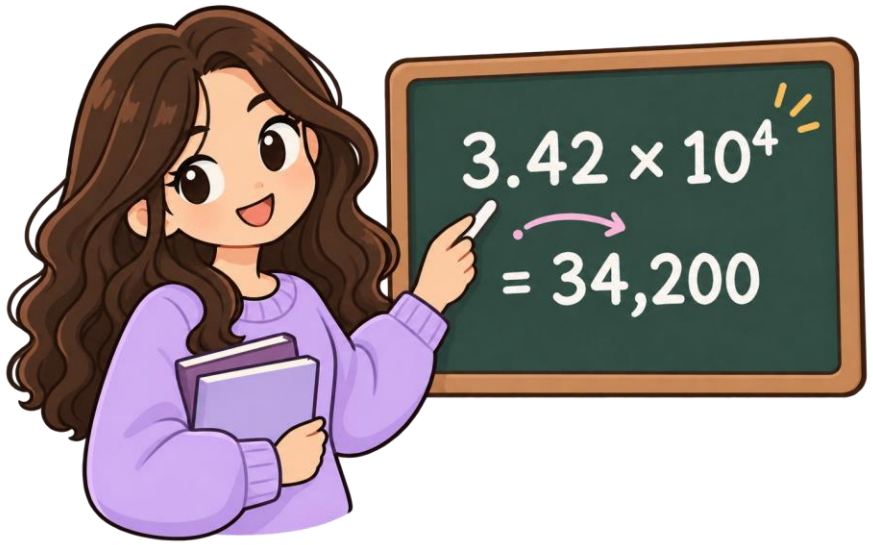
If the p-value is 0.01 or less:

- It's **unlikely** to obtain this sample mean (at the **99%** level)
- There **is a significant difference** between the sample mean and population parameter
- **Reject** the null hypothesis

Important Notes:

- Posit Cloud defaults to **95%** as the confidence level and .05 as the alpha level
- The p-value can be a very small proportions
- Posit Cloud **may use scientific notation** to denote the p-value

Reading Scientific Notations



Powers of 10 (positive exponents)

10	=	10^1	= e+1
100	=	10^2	= e+2
1000	=	10^3	= e+3

Powers of 10 (negative exponents)

0.1	=	10^{-1}	= e-1
0.01	=	10^{-2}	= e-2
0.001	=	10^{-3}	= e-3

More Examples

- **8334** = 8.334×10^3 = $8.334 \times$ e+3 = 8.334e+3
- **6.189 e-4** = 6.189×10^{-4} = 6.189×0.0001 = 0.0006189
- **0.05** = 5×10^{-2} = 5e-2 (so any number times e-3, e-4, e-5, e-6… will all be smaller than 0.05)
- **0.01** = 1×10^{-2} = 1e-2 (so any number times e-3, e-4, e-5, e-6… will all be smaller than 0.01)

Moving the Decimal Point

Positive exponent → move the decimal point **to the right**

7.74521×10^5
= 7.74521
(move 5 places to the right)
= **774 521**

- The exponent is positive (5), so move the decimal point 5 places **to the right**.
- No zeros are needed to fill empty gaps.

Negative exponent → move the decimal point **to the left**

6×10^{-3}
= 6.
(move 3 places to the left)
= **0.006**

- The exponent is negative (3), so move the decimal point 3 places **to the left**.
- Add in zeros to fill the empty gaps.

Same Example Using Posit Cloud



```
15
16 ```{r}
17 gss <- read_excel('GSS2006.xlsx', sheet='GSS 2006')
18 ```
19
20 ```{r}
21 nrow(gss)
22 mean(gss$age)
23 sd(gss$age)
24 ```
```

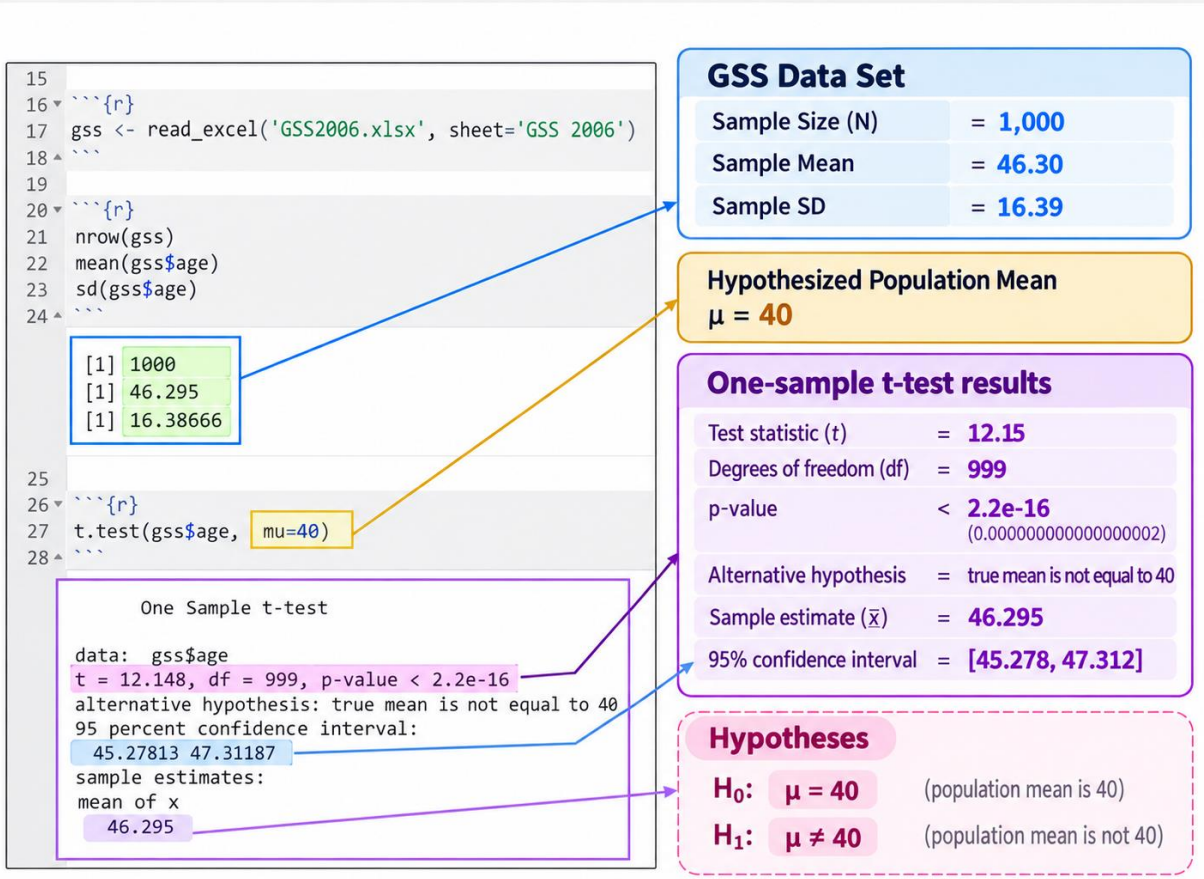
```
[1] 1000
[1] 46.295
[1] 16.38666
```

```
25
26 ```{r}
27 t.test(gss$age, mu=40)
28 ```
```

```
        One Sample t-test

data:  gss$age
t = 12.148, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 40
95 percent confidence interval:
 45.27813 47.31187
sample estimates:
mean of x
   46.295
```

Same Example Using Posit Cloud



Same Example Using Posit Cloud

The sample average age is 46.30 years old.

The probability of obtaining this sample mean if the true population average age is 40 is less than 0.0000000000000002.

In other words, the probability of error if we rejected the null hypothesis is .0000000000000002.

```
One Sample t-test

data:  gss$age
t = 12.148, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 40
95 percent confidence interval:
 45.27813 47.31187
sample estimates:
mean of x
   46.295
```

At the 0.05 significance (alpha) level, we *reject the null hypothesis* and conclude that there is a statistically significant difference between the sample average age of 46 and the hypothesized population average age of 40.

In other words, the true population average age is likely NOT 40.

Discussion for Homework Assignment 7

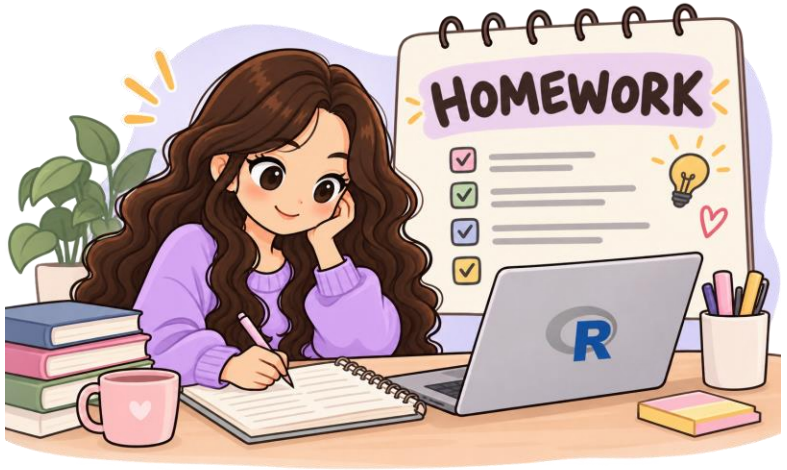
Project Link: https://posit.cloud/content/7102503

Use the link to the Project

Save it as a permanent project in your account

Rename the project, Open the markdown file and resize panes

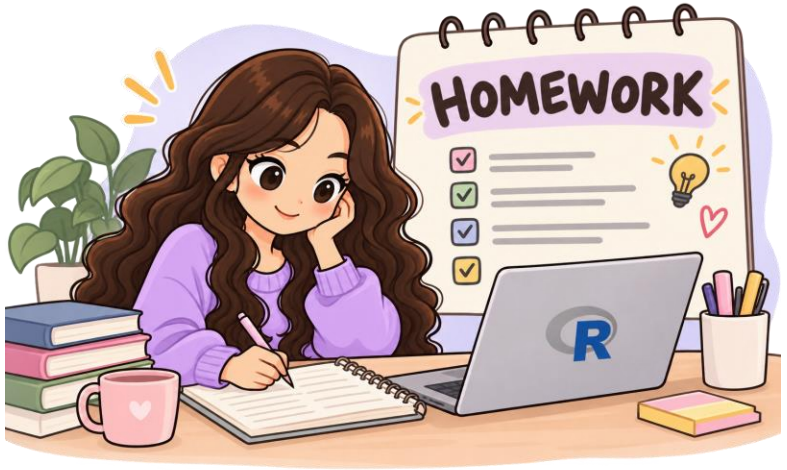
Read all comments and follow the steps in the markdown file:



- First code chunk will load two packages
- Second code chunk will load and inspect the GSS data frame
- Third code chunk will **ask you to write some code** to calculate some statistics for the variable hrs1
- Fourth code chunk will calculate a one sample t-test

Discussion for Homework Assignment 7

Project Link: https://posit.cloud/content/7102503



Change the author (line 3) and date (line 4) of the markdown file

Knit the Markdown file and save it to your computer/device

Open the Assignment in Brightspace

Answer the questions in Brightspace

Submit the knitted file in Brightspace

Due on November 1st at 11:59pm

Q1. Hypothesis Testing by hand (6 steps), hypothesized population mean = 40.

Q2. Hypothesis Testing in Posit Cloud, hypothesized population mean = 43.